Disaster Recovery Strategies

pay for the recovery speed you need

Watch the 4-min video

In one line: the four classic DR strategies (backup and restore, pilot light, warm standby, active-active) trade money for recovery speed and data loss; choose per system from its RTO and RPO, and only trust the one you have actually rehearsed.

1 A whole region goes dark

Not a server, not a zone: the whole region. Do you recover in ten minutes or in two days? That was decided months ago, when you chose a **disaster recovery** strategy and how much to pay for it. Zone redundancy (lesson 10) doesn't help when the entire region, its control plane or your account in it is unavailable.

2 Four strategies



Figure 1. From cheapest and slowest to most expensive and fastest. Each step keeps more of production running in the second region.

Strategy	In the DR region	RTO	RPO	Cost
Backup + restore	backups only; rebuild infra, then restore data	hours to a day	since last backup	lowest
Pilot light	replicated data, infra as code, app servers off	tens of minutes	seconds–minutes	low
Warm standby	scaled-down but working copy of production	minutes	seconds–minutes	medium
Active-active (multi-site)	full production, serving traffic	near zero	near zero	highest

Backup and restore is cheap because nothing runs, but you rebuild everything: infrastructure first, then hours of data restore. **Pilot light** keeps the core (data) always on and small; you start and scale the app tier during the disaster. **Warm standby** can take some traffic at once and is scaled up during failover. **Active-active** fails over almost instantly, but writes in two regions must be reconciled (lesson 15), which makes it the most complex option. These names and ranges follow AWS's DR whitepaper; other clouds describe the same ladder.

3 A plan you haven't tested is a hope

- Run **DR drills** that really fail over, measure the time, and compare it with your RTO and RPO (lesson 13).
- Keep **runbooks** current and **automate** the steps: during a real disaster people are stressed and make mistakes.
- Check the things people forget: DNS, secrets and certificates, quotas in the DR region, container images and third-party allow-lists.
- Keep backups in a separate account too, so the disaster can't be a deleted or compromised account (lesson 14).

WORKED EXAMPLE: PILOT LIGHT FOR AN INTERNAL EXPENSE APP

The app can tolerate **an hour** of downtime. A warm standby would roughly double its cost, so the team picks **pilot light**: the database replicates to a second region, and the app's infrastructure exists as **Terraform code** with **zero app servers** running. In a regional outage, a runbook launches the servers and switches DNS; the app is back in about **30 minutes**, inside its one-hour target, at a few percent of the cost of a second production. They drill it every quarter.

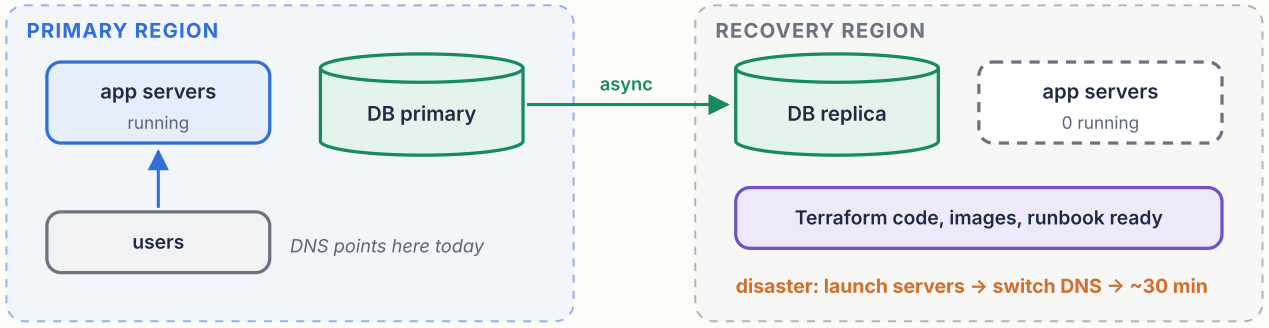


Figure 2. Pilot light: only the data is live in the recovery region; everything else is ready to start.

COMMON MISTAKES

- One strategy for every system, instead of matching each system's RTO/RPO.
- A DR region that depends on the primary region (its control plane, its DNS, its CI).
- Capacity assumptions: the DR region may not have quota or instances when everyone fails over.
- Never drilling, so the first real failover is also the first test.

4 Key terms

Disaster recovery (DR): restoring service after a large-scale loss

Backup and restore: rebuild from backups; cheapest, slowest

Pilot light: data live, app tier off until needed

Warm standby: small working copy, scaled up on failover

Active-active: every region serves traffic all the time

DR drill: a real, measured failover rehearsal

5 Interview questions

> **Q1. Name the four classic DR strategies.** BASIC
> Backup and restore, pilot light, warm standby and multi-site active-active, in order of rising cost and falling recovery time.

Q2. What's the difference between pilot light and warm standby? BASIC

Pilot light replicates data but keeps app servers off; warm standby runs a small working copy that can take traffic at once and is scaled up.

Q3. Compare DR strategies by cost and recovery time. SCENARIO

Backup and restore: cheapest, hours to a day. Pilot light: tens of minutes. Warm standby: minutes. Active-active: most expensive and complex, near-instant failover.

Q4. An internal app can be down for an hour. Which strategy? SCENARIO

Pilot light: replicate the data, keep infra as code, launch servers and switch DNS in about 30 minutes, at a fraction of a warm standby's cost.

Q5. How do you know your DR plan works? TRICKY

Run regular drills that really fail over, measure recovery against RTO and RPO, and fix the missing steps.

Q6. Why isn't active-active always the answer? TRICKY

It's the most expensive and complex: it needs multi-region data replication, conflict handling and capacity for full load in each region, which many systems don't justify.

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